

Question #1 of 34

Question ID: 1473464

Jill Sebelius, editor-in-chief of a monthly interest-rate newsletter uses the following model to forecast short-term interest rates:

$$dr = a(b - r) dt + \sigma\sqrt{r}dz$$

For the current newsletter, Sebelius has issued the following expectations:

$$a=0.40, b = 3\%, r = 2\%.$$

According to the model used by Sebelius, volatility in the short-term interest rate is *most likely*.

- A)** independent of the current level of the short-term interest rate.
- B)** negatively related to the current level of the short-term interest rate.
- C)** positively related to the current level of the short-term interest rate.

Dan Laske, CFA, recently joined Axes Bank as a bond trader. Laske is reviewing bank's interest rate models and meets with Sheila Jensen, a senior portfolio manager at the bank.

Jensen makes the following statements:

- Statement 1: I don't want any random noise in the short-term rate forecasts.
- Statement 2: I like interest models to incorporate mean reversion of rates.
- Statement 3: I prefer that the interest rate volatility varies with the level of interest rates.

Laske finally decides on using the Gauss+ model and asks Jensen about her opinion.

Jensen makes the following observations about the Gauss+ model:

- Observation 1: Gauss+ models are primarily used for forecasting long-term rates, while equilibrium models are used for short-term rates.
- Observation 2: Gauss+ models assume that the long-term rates are mean reverting, while the central bank actions determine the short-term rates.

Question #2 - 5 of 34

Question ID: 1494708

Based on Jensen's Statement 1, which model is *most appropriate*?

- A)** The Kalotay-Williams-Fabozzi (KWF) model.
 - B)** A Gauss+ multifactor model.
 - C)** The Ho-Lee model.
-

Question #3 - 5 of 34

Question ID: 1494709

Based on Jensen's Statement 2, which model is *least appropriate*?

- A)** The Vasicek model.
 - B)** The Cox-Ingersoll-Ross (CIR) model.
 - C)** The Ho-Lee model.
-

Question #4 - 5 of 34

Question ID: 1494710

Based on Jensen's Statement 3, which model is *most appropriate*?

- A)** The Cox-Ingersoll-Ross (CIR) model.
 - B)** The Kalotay-Williams-Fabozzi (KWF) model.
 - C)** The Vasicek model.
-

Question #5 - 5 of 34

Question ID: 1494711

Regarding Jensen's observations:

- A)** both observations are incorrect.
 - B)** only Observation 2 is correct.
 - C)** only Observation 1 is correct.
-

Question #6 of 34

Question ID: 1473436

The government bond spot rate curve is given below:

Maturity (years)	Spot rate
0.5	1.25%
1.0	1.30%
1.5	1.80%
2.0	2.00%
2.5	2.20%
3.0	2.25%
3.5	2.28%
4.0	2.30%

Compute the issue price of a 3-year, 3% semiannual coupon government bond with a par value of \$100.

- A)** \$102.15
 - B)** \$104.09
 - C)** \$102.20
-

Question #7 of 34

Question ID: 1473460

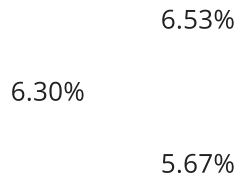
Relative to the binomial model, Monte Carlo method is *most likely*:

- A)** less flexible in forcing interest rates to mean revert.
 - B)** more flexible as it does not need a volatility estimate.
 - C)** more suitable when valuing securities whose cash flows are interest rate path dependent.
-

Question #8 of 34

Question ID: 1473447

Using the following interest rate tree of semiannual interest rates what is the value of an option free semiannual bond that has one year remaining to maturity and has a 6% coupon rate?



- A) 99.81.
 - B) 97.53.
 - C) 98.52.
-

Question #9 of 34

Question ID: 1473459

A binomial model or any other model that uses the backward induction method cannot be used to value a mortgage-backed security (MBS) because:

- A) the cash flows for an MBS only depend on the current rate, not the path that rates have followed.
 - B) the cash flows for the MBS are dependent upon the path that interest rates follow.
 - C) the prepayments occur linearly over the life of an interest rate trend (either up or down).
-

Question #10 of 34

Question ID: 1473453

For a 3-year, semiannual coupon payment bond, the number of interest rate paths that would be generated using the pathwise valuation is *closest* to:

- A) 4
 - B) 64
 - C) 32
-

Sam Roit, CFA, has collected the following information on the par rate curve, spot rates, and forward rates to generate a binomial interest rate tree consistent with this data.

Maturity	Par Rate	Spot Rate
1	5%	5.000%
2	6%	6.030%
3	7%	7.097%

The binomial tree generated is shown below (one year forward rates) assuming a volatility level of 10%:

0	1	2
5%	7.7099%	C
	A	9.2625%
		B

Riot also generated another tree using the same spot rates but this time assuming a volatility level of 20% as shown below:

0	1	2
5%	8.9480%	13.8180%
	5.9980%	9.2625%
		6.2088%

The one-year forward rate represented by 'A' is closest to:

- A) 6.3123%
- B) 5.4223%
- C) 6.7732%

A 3-year, 3% annual pay, \$100 par bond is valued using pathwise valuation. The interest rate paths are provided below:

Path	Year 1	Year 2	Year 3
1	2%	2.8050%	4.0787%
2	2%	2.8050%	3.0216%
3	2%	2.0780%	3.0216%
4	2%	2.0780%	2.2384%

The value of the bond in path 1 is *closest* to:

- A) \$101.88
 - B) \$100.18
 - C) \$98.77
-

Question #13 of 34

Question ID: 1473455

A 3-year, 3% annual pay, \$100 par bond is valued using pathwise valuation. The interest rate paths are provided below:

Path	Year 1	Year 2	Year 3
1	2%	2.8050%	4.0787%
2	2%	2.8050%	3.0216%
3	2%	2.0780%	3.0216%
4	2%	2.0780%	2.2384%

The value of the bond in path 4 is *closest* to:

- A) \$102.58
 - B) \$101.88
 - C) \$100.02
-

Question #14 of 34

Question ID: 1473456

A 3-year, 3% annual pay, \$100 par bond is valued using pathwise valuation. The interest rate paths are provided below:

Path	Year 1	Year 2	Year 3
1	2%	2.8050%	4.0787%
2	2%	2.8050%	3.0216%
3	2%	2.0780%	3.0216%
4	2%	2.0780%	2.2384%

The value of the bond in path 2 is *closest* to:

- A) \$100.88
 - B) \$101.15
 - C) \$102.72
-

Question #15 of 34

Question ID: 1473443

With respect to interest rate models, backward induction refers to determining:

- A) convexity from duration.
 - B) the current value of a bond based on possible final values of the bond.
 - C) one portion of the yield curve from another portion.
-

Question #16 of 34

Question ID: 1473444

Which of the following is a *correct* statement concerning the backward induction technique used within the binomial interest rate tree framework? From the maturity date of a bond:

- A) the corresponding interest rates are weighted by the bond's duration to discount the value of the bond.
- B) the corresponding interest rates and interest rate probabilities are used to discount the value of the bond.

C) a deterministic interest rate path is used to discount the value of the bond.

Question #17 of 34

Question ID: 1473450

Sam Roit, CFA, has collected the following information on the par rate curve, spot rates, and forward rates to generate a binomial interest rate tree consistent with this data.

Maturity	Par Rate	Spot Rate
1	5%	5.000%
2	6%	6.030%
3	7%	7.097%

The binomial tree generated is shown below (one year forward rates) assuming a volatility level of 10%:

0	1	2
5%	7.7099%	C
	A	9.2625%
		B

Roit also generated another tree using the same spot rates but this time assuming a volatility level of 20% as shown below:

0	1	2
5%	8.9480%	13.8180%
	5.9980%	9.2625%
		6.2088%

The one-year forward rate represented by 'B' is closest to:

A) 8.7732%

B) 7.5835%

C) 7.4223%

Question #18 of 34

Question ID: 1473462

Jill Sebelius, editor-in-chief of a monthly interest-rate newsletter uses the following model to forecast short-term interest rates:

$$dr = a(b - r) dt + \sigma\sqrt{r}dz$$

For the current newsletter, Sebelius has issued the following expectations:

$$a=0.40, b = 3\%, r = 2\%.$$

Based on Sebelius's estimates, over a sufficiently long period of time, the expected value of the short-term interest rate is *closest* to:

- A) 3%
 - B) 2.4%
 - C) 2%
-

Question #19 of 34

Question ID: 1473435

The process of stripping is *most* likely to be used to earn arbitrage profits in a situation where:

- A) a portfolio of treasury strips is trading for a lower price than an intact treasury bond.
 - B) one treasury bond trades at a lower price than another treasury bond with identical characteristics.
 - C) Security valuations are not consistent with the value additivity principle.
-

Question #20 of 34

Question ID: 1473437

Government par curve is provided below:

Maturity (years)	Par rate
1	5.0%
2	6.0%
3	6.5%
4	7.0%

The value of a 4-year, 5% annual pay, \$100 par government bond is closest to:

- A) \$98.49
 - B) \$93.15
 - C) \$101.12
-

Question #21 of 34

Question ID: 1473438

Tim Brospace is generating a binomial interest rate tree assuming a volatility of 15%. Current 1-year spot rate is 5%. The 1-year forward rate in the second year is either a low estimate of 5.250% or a high estimate of 7.087%. The middle 1-year forward rate in year three is estimated at 6.25%. The upper node 1-year forward rate in year three is *closest* to:

- A) 7.747%
 - B) 6.445%
 - C) 8.437%
-

Question #22 of 34

Question ID: 1508672

Which of the following choices is *least likely* a property of a binomial interest rate tree?

- A) Non-negative interest rates.
 - B) Adjacent forward rates in a nodal period are one standard deviation apart.
 - C) Higher volatility at higher rates.
-

Question #23 of 34

Question ID: 1473461

Currently the term structure of interest rate is downward sloping. Which of the following models *most accurately* describe the current term structure?

- A) Vasicek model.
 - B) Cox-Ingersoll-Ross model.
 - C) Ho-Lee model.
-

Question #24 of 34

Question ID: 1508671

Which of the following choices is *least likely* a property of a binomial interest rate tree?

- A) Mean reversion of interest rates.
 - B) Non-negative interest rates.
 - C) Higher volatility at higher rates.
-

Question #25 of 34

Question ID: 1473463

Jill Sebelius, editor-in-chief of a monthly interest-rate newsletter uses the following model to forecast short-term interest rates:

$$dr = a(b - r) dt + \sigma\sqrt{r}dz$$

For the current newsletter, Sebelius has issued the following expectations:

$$a=0.40, b = 3\%, r = 2\%.$$

Sebelius's model is *most accurately* described as the:

- A) Vasicek model.
 - B) Ho-Lee model.
 - C) Cox-Ingersoll-Ross model.
-

Question #26 of 34

Question ID: 1473448

Sam Roit, CFA, has collected the following information on the par rate curve, spot rates, and forward rates to generate a binomial interest rate tree consistent with this data.

Maturity	Par Rate	Spot Rate
1	5%	5.000%
2	6%	6.030%
3	7%	7.097%

The binomial tree generated is shown below (one year forward rates) assuming a volatility level of 10%:

0	1	2
5%	7.7099%	C
	A	9.2625%
		B

Riot also generated another tree using the same spot rates but this time assuming a volatility level of 20% as shown below:

0	1	2
5%	8.9480%	13.8180%
	5.9980%	9.2625%
		6.2088%

The one-year forward rate represented by 'C' is closest to:

- A) 8.7732%
- B) 7.4223%
- C) 11.3132%

Question #27 of 34

Question ID: 1473458

Increasing the number of paths generated in a Monte Carlo simulation is *most likely* to increase the:

- A) statistical accuracy of the estimated value.
 - B) fundamental accuracy of the estimated value.
 - C) utility of the model.
-

Question #28 of 34

Question ID: 1473445

A bond with a 10% annual coupon will mature in two years at par value. The current one-year spot rate is 8.5%. For the second year, the yield volatility model forecasts that the one-year rate will be either 8% or 9%. Using a binomial interest rate tree, what is the current price?

- A) 101.837.
 - B) 103.572.
 - C) 102.659.
-

Question #29 of 34

Question ID: 1473439

Tim Brospack is generating a binomial interest rate tree assuming a volatility of 15%. Current 1-year spot rate is 5%. The 1-year forward rate in the second year is either a low estimate of 5.250% or a high estimate of 7.087%. The middle 1-year forward rate in year three is estimated at 6.25%. The lower node 1-year forward rate in year three is *closest* to:

- A) 6.747%
 - B) 4.63%
 - C) 5.342%
-

Question #30 of 34

Question ID: 1473451

Sam Roit, CFA, has collected the following information on the par rate curve, spot rates, and forward rates to generate a binomial interest rate tree consistent with this data.

Maturity	Par Rate	Spot Rate
1	5%	5.000%
2	6%	6.030%
3	7%	7.097%

The binomial tree generated is shown below (one year forward rates) assuming a volatility level of 10%:

0	1	2
5%	7.7099%	C
	A	9.2625%
		B

Roit also generated another tree using the same spot rates but this time assuming a volatility level of 20% as shown below:

0	1	2
5%	8.9480%	13.8180%
	5.9980%	9.2625%
		6.2088%

Is the binomial tree using the 20% volatility assumption calibrated properly?

- A)** The tree is not calibrated properly because it is not consistent with market prices.
- B)** The tree is calibrated properly.
- C)** The tree is not calibrated properly because adjacent nodes are not appropriate standard deviations apart.

Why is the backward induction methodology used to value a bond rather than a forward induction scheme?

- A) The convexity of a bond changes over time.
 - B) The price of the bond is known at maturity.
 - C) Future interest rate changes are difficult to forecast.
-

Question #32 of 34

Question ID: 1473446

Using the following interest rate tree of semiannual interest rates what is the value of an option free bond that has one year remaining to maturity and has 5% coupon rate with semi-annual coupon payments.

Today	6 Months
	7.30%
6.20%	
	5.90%

- A) 97.53.
 - B) 98.98.
 - C) 98.67.
-

Question #33 of 34

Question ID: 1473452

Suppose that we calculate the value of an option-free, fixed-rate coupon bond, discounting the cash flows using two methods:

- I. the zero-coupon yield curve.
- II. an arbitrage-free binomial lattice.

Compared to the first methodology, the second method is expected to produce:

- A) the same value.
 - B) a lower value if the bond carries a coupon higher than the corresponding benchmark bond.
 - C) a higher value in the presence of volatility.
-

Question #34 of 34

Question ID: 1473454

A 3-year, 3% annual pay, \$100 par bond is valued using pathwise valuation. The interest rate paths are provided below:

Path	Year 1	Year 2	Year 3
1	2%	2.8050%	4.0787%
2	2%	2.8050%	3.0216%
3	2%	2.0780%	3.0216%
4	2%	2.0780%	2.2384%

The value of the bond in path 3 is closest to:

- A)** \$99.88
- B)** \$100.02
- C)** \$101.85